

Arduino Intro & Digital!

What is Arduino

- An **open-source** platform ?
- Based on an easy-to-use hardware and software
- User friendly Development Environment (IDE)
- Created for teaching electronics to artists and designers.
- Low cost (~ 400 INR)
- A Microcontroller?



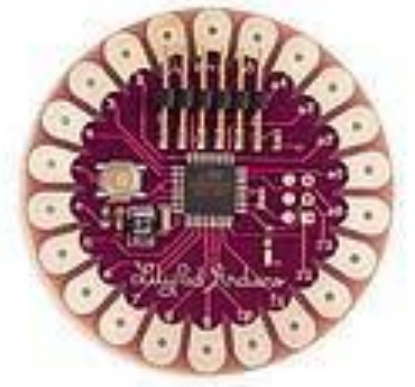
Arduino Uno



Arduino Leonardo



Arduino Mega 2560



Arduino LilyPad



Arduino Mega ADK



Arduino Fio



Arduino Ethernet



Arduino Pro



Arduino BT



Arduino Nano



Arduino Mini



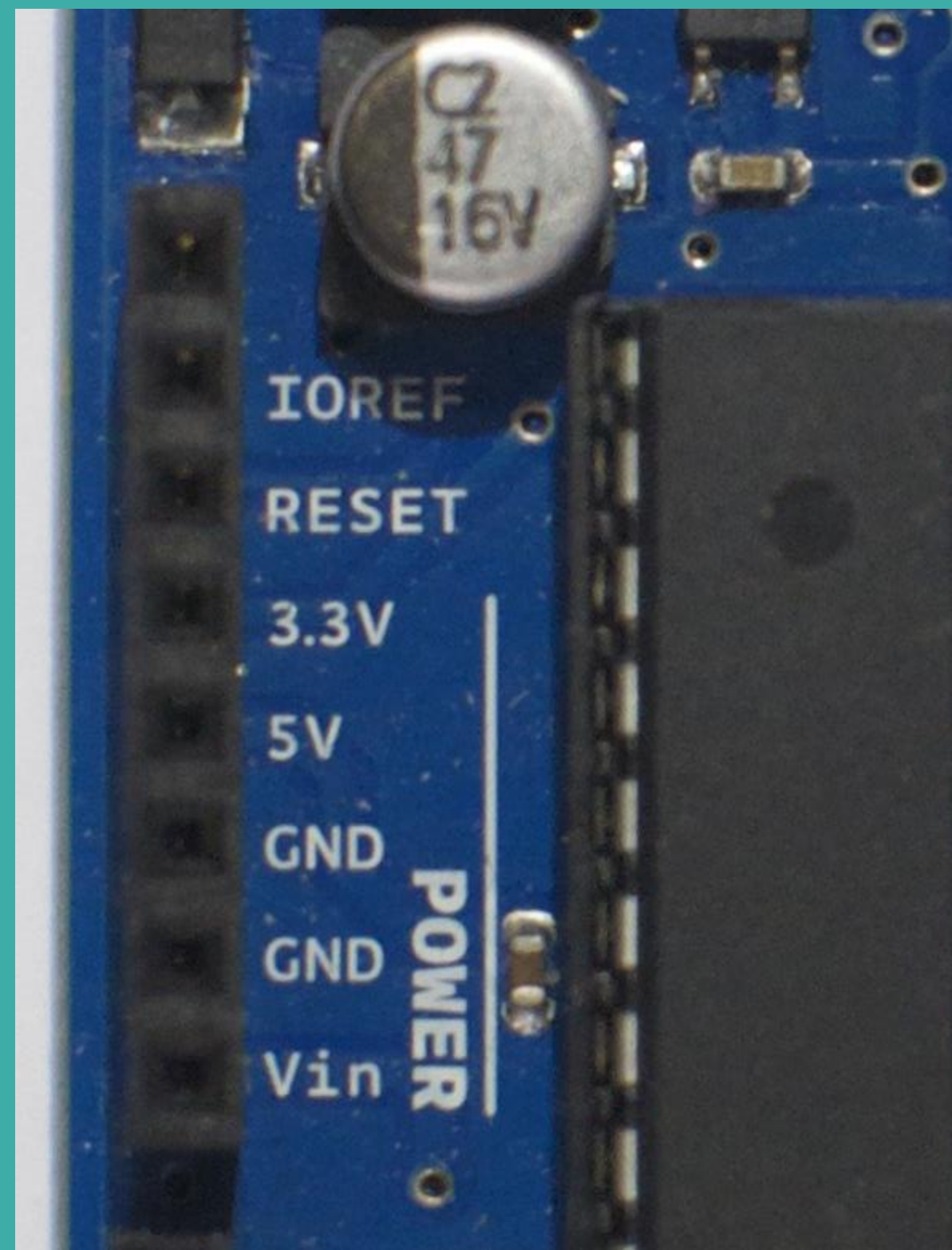
Arduino Pro Mini

Board Overview

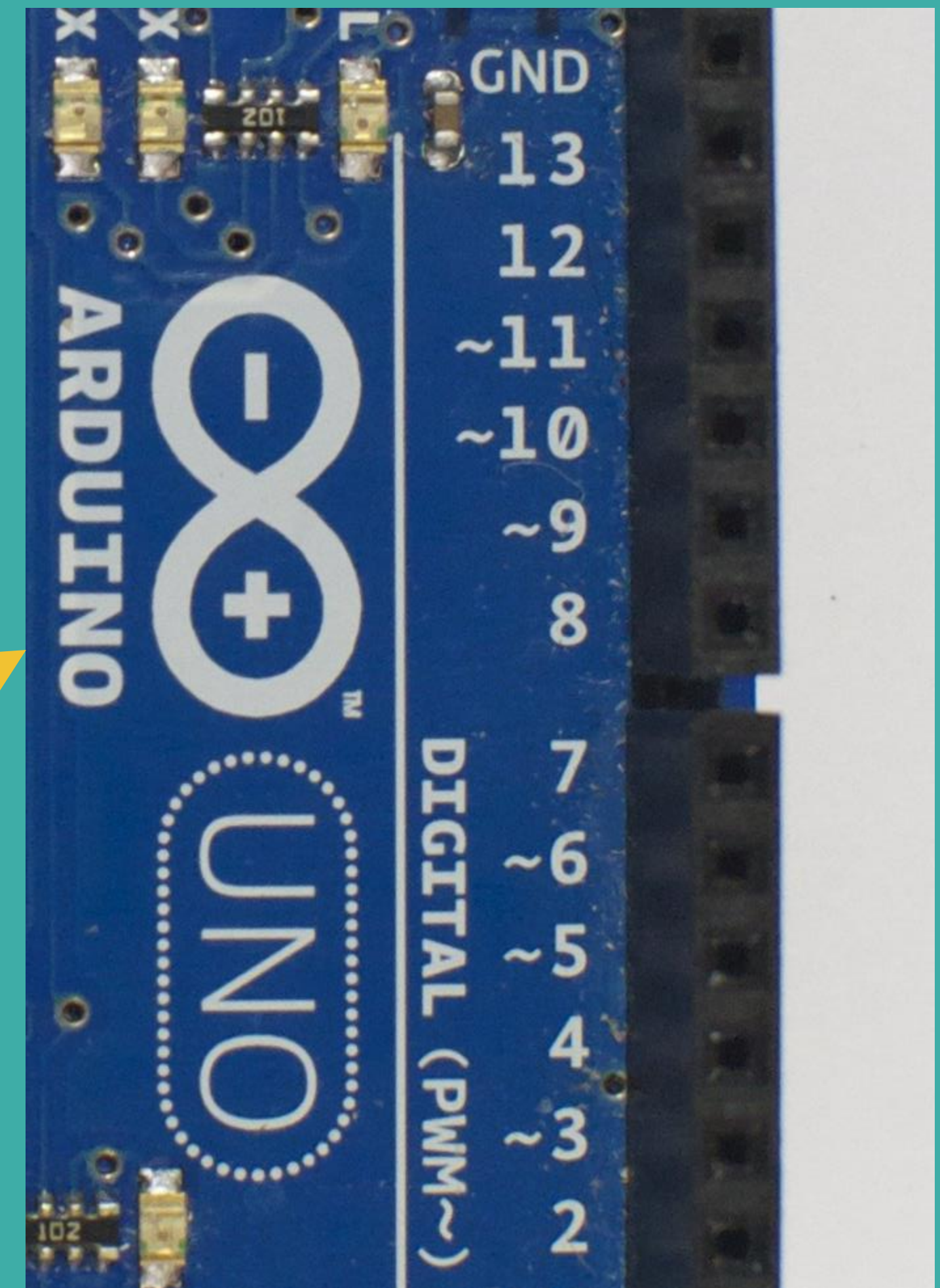
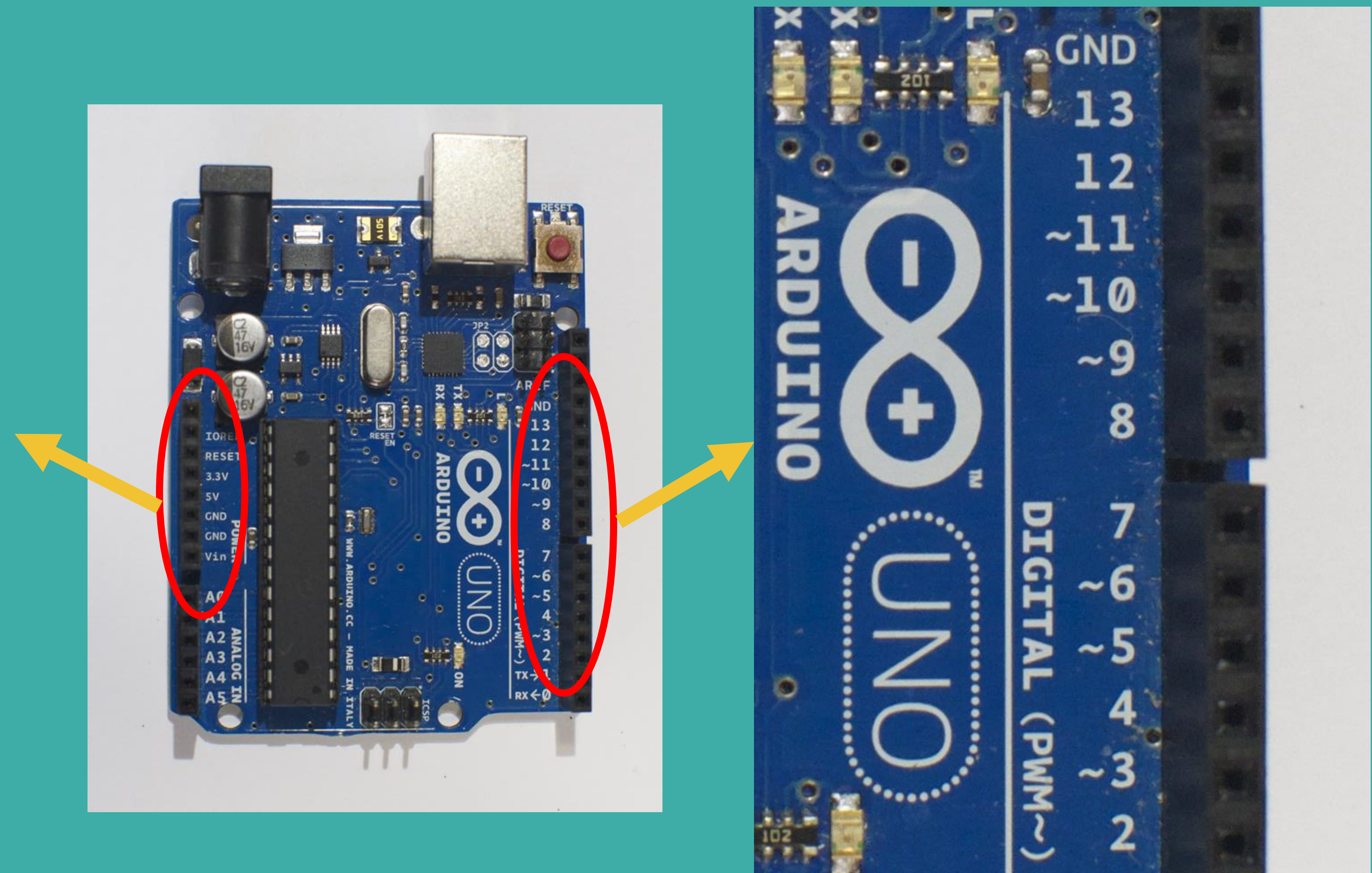
- Microcontroller
- Crystal Oscillator
- Power Pins
- Analog Pins
- Digital Pins
- Tx-Rx Pins
- Reset Button

Power & Digital Pins

Power Pins



Digital Pins



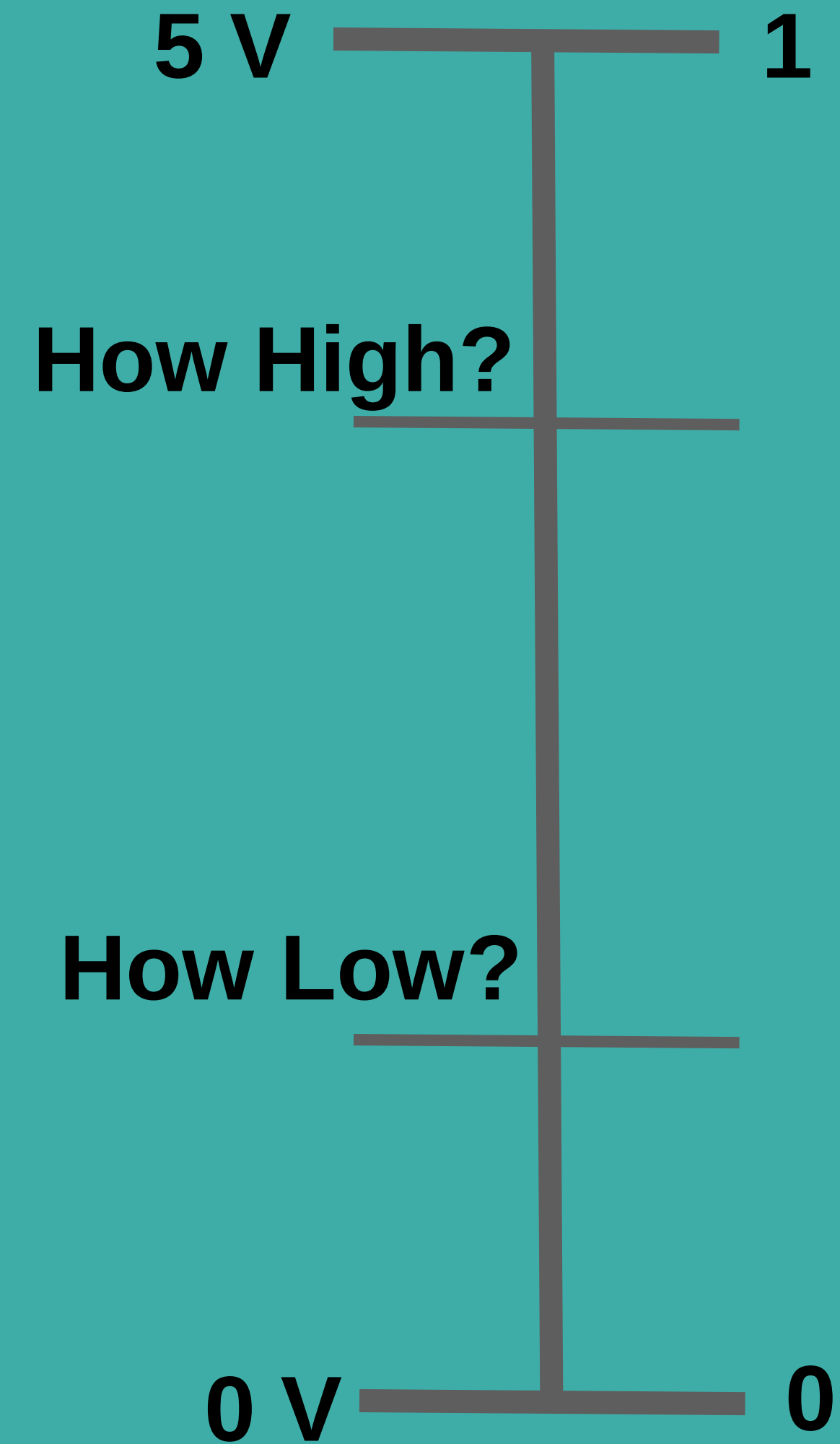
Digital Pins

Logic Levels:

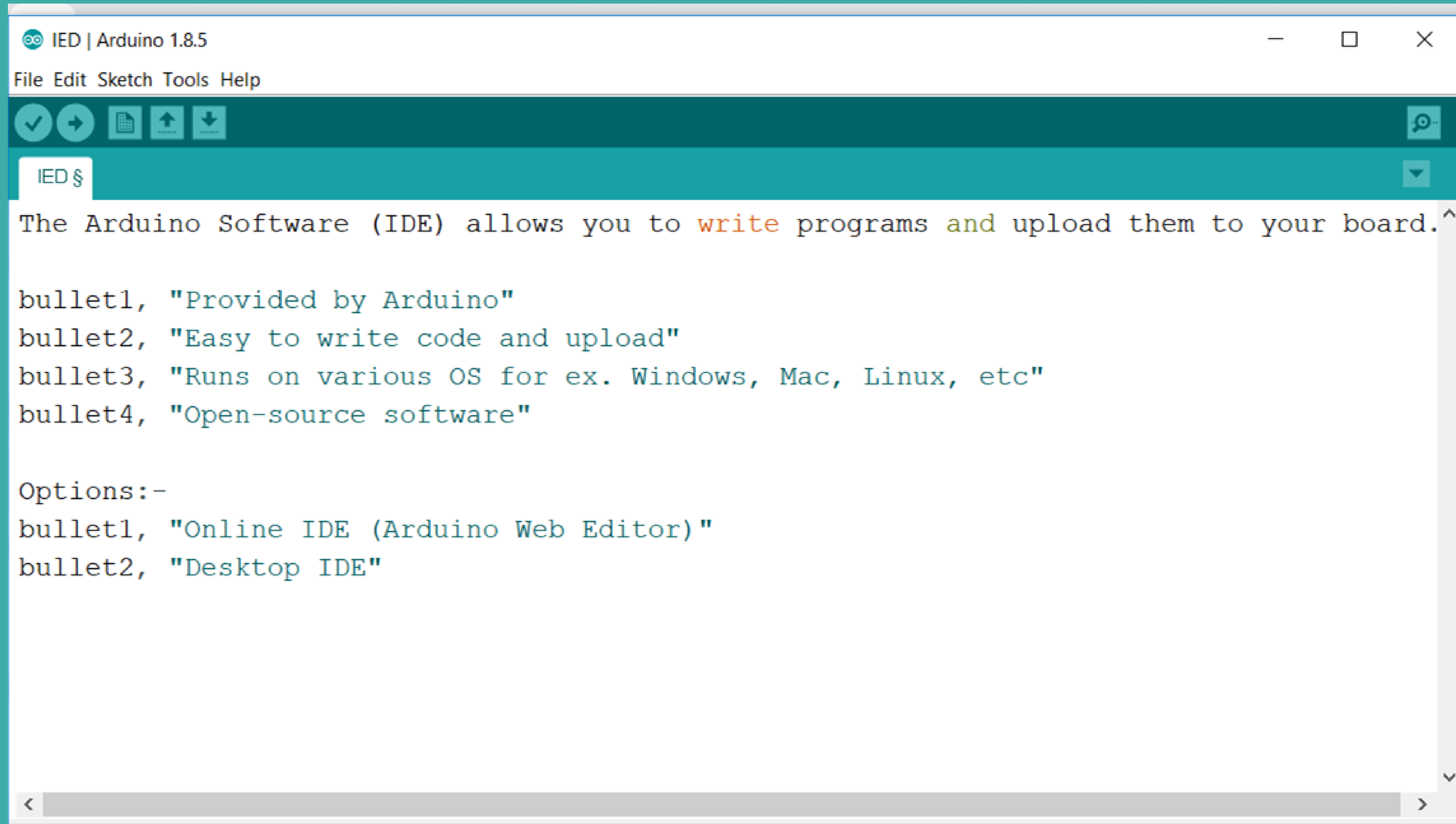
- 0
- 1

Logic Families:

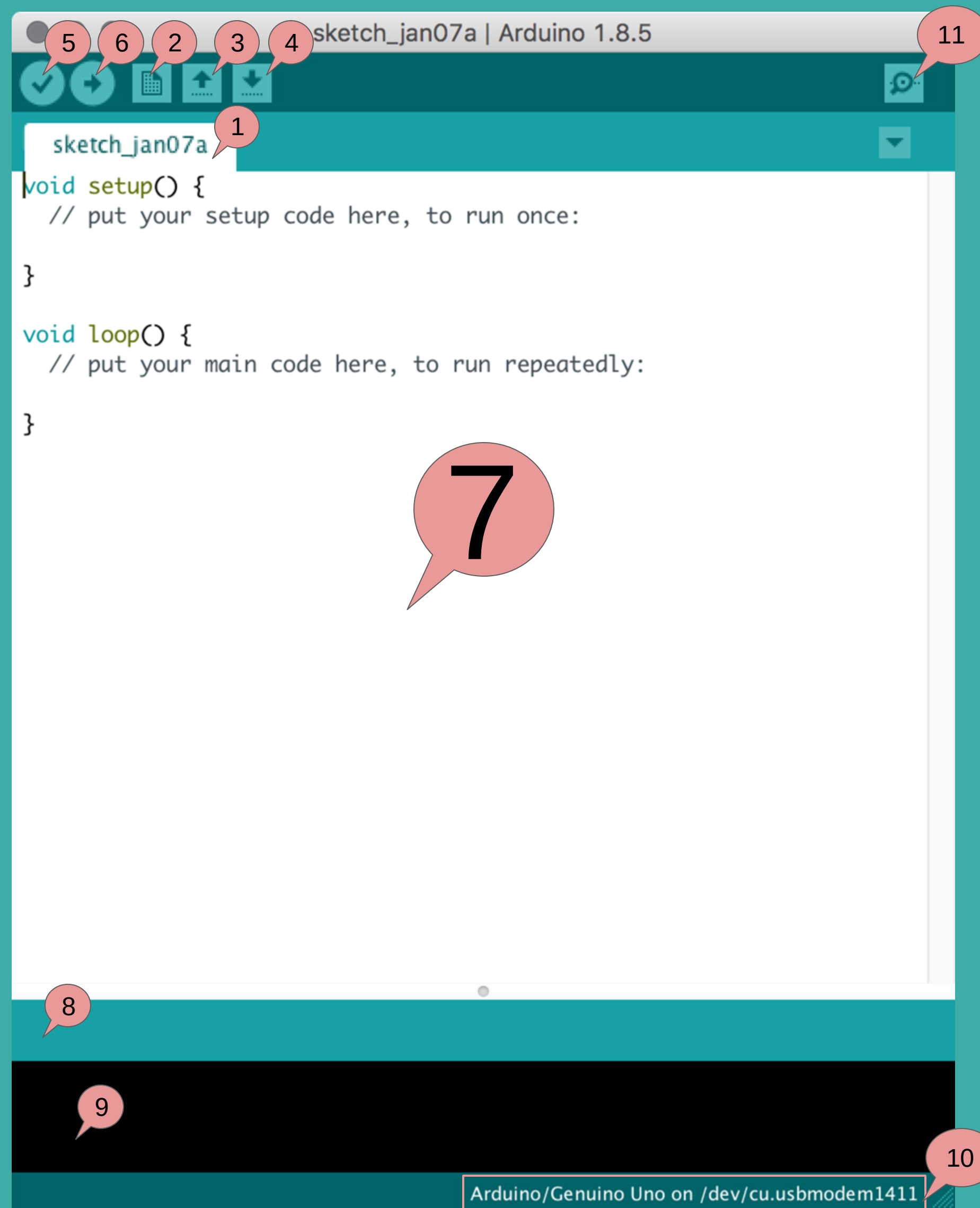
- CMOS
- TTL
- ECL
- RTL
- DTL
- and A lot more



Arduino IDE



Arduino IDE



1. **Sketch Name:** This shows the name of the sketch you are currently working on.
2. **New:** This buttons opens up a new code window tab.
3. **Open:** This button will let you open up an existing sketch.
4. **Save:** This saves the currently active sketch.

1. **Verify:** Compiles and approves your code. It will catch errors in syntax (like missing semicolons or parenthesis).
2. **Upload:** Sends your code to the Arduino board. When you click it, you should see the lights on your board blink rapidly.

1. **Code Area:** This is the area where you compose the code for your sketch.
2. **Message Area:** This is where the IDE tells you if there were any errors in your code.
3. **Text Console:** The text console shows complete error messages. When debugging, the text console is very useful.

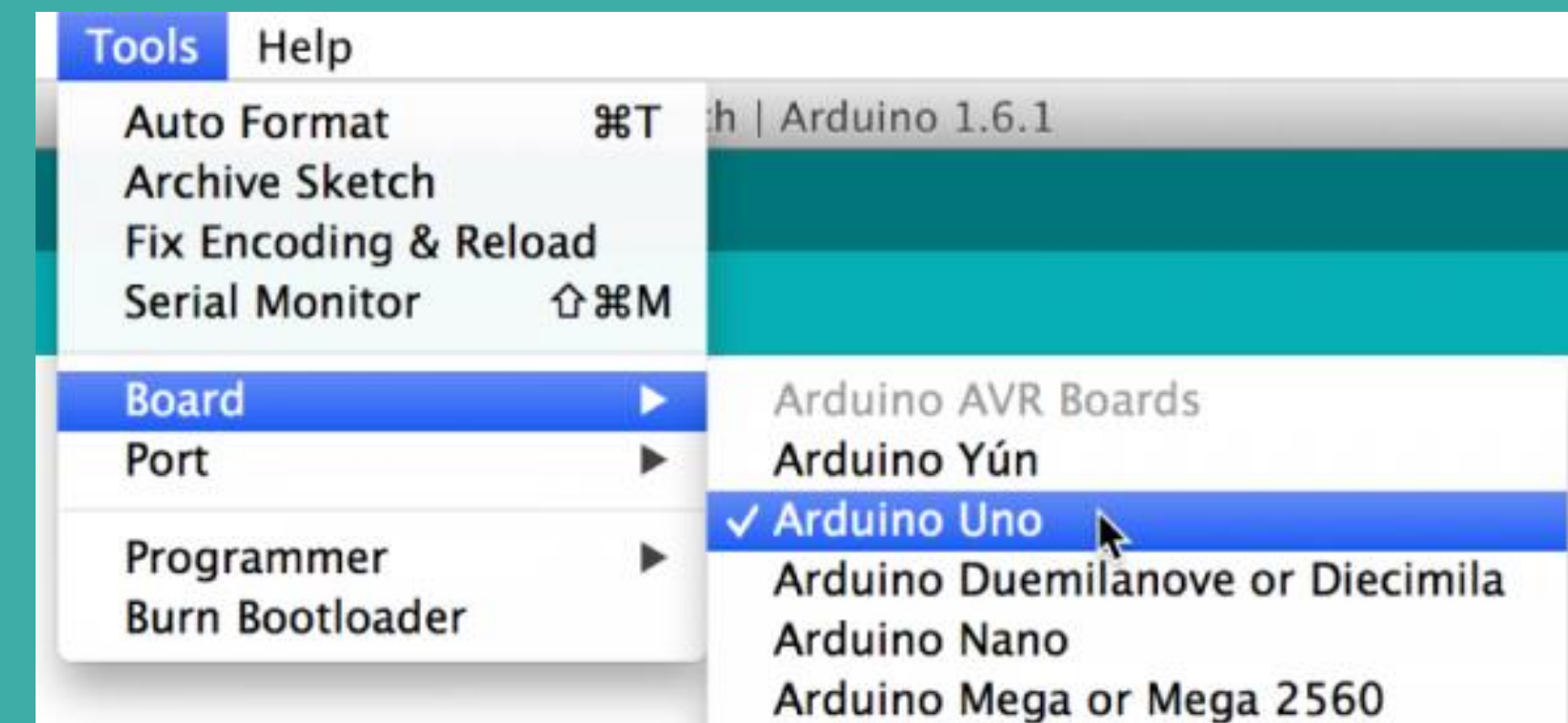
1. **Board and Serial Port:** Shows you what board and the serial port selections
2. **Serial Monitor:** This will open a window that displays any serial information your Arduino is transmitting. It is very useful for debugging.

Board and Port Selection

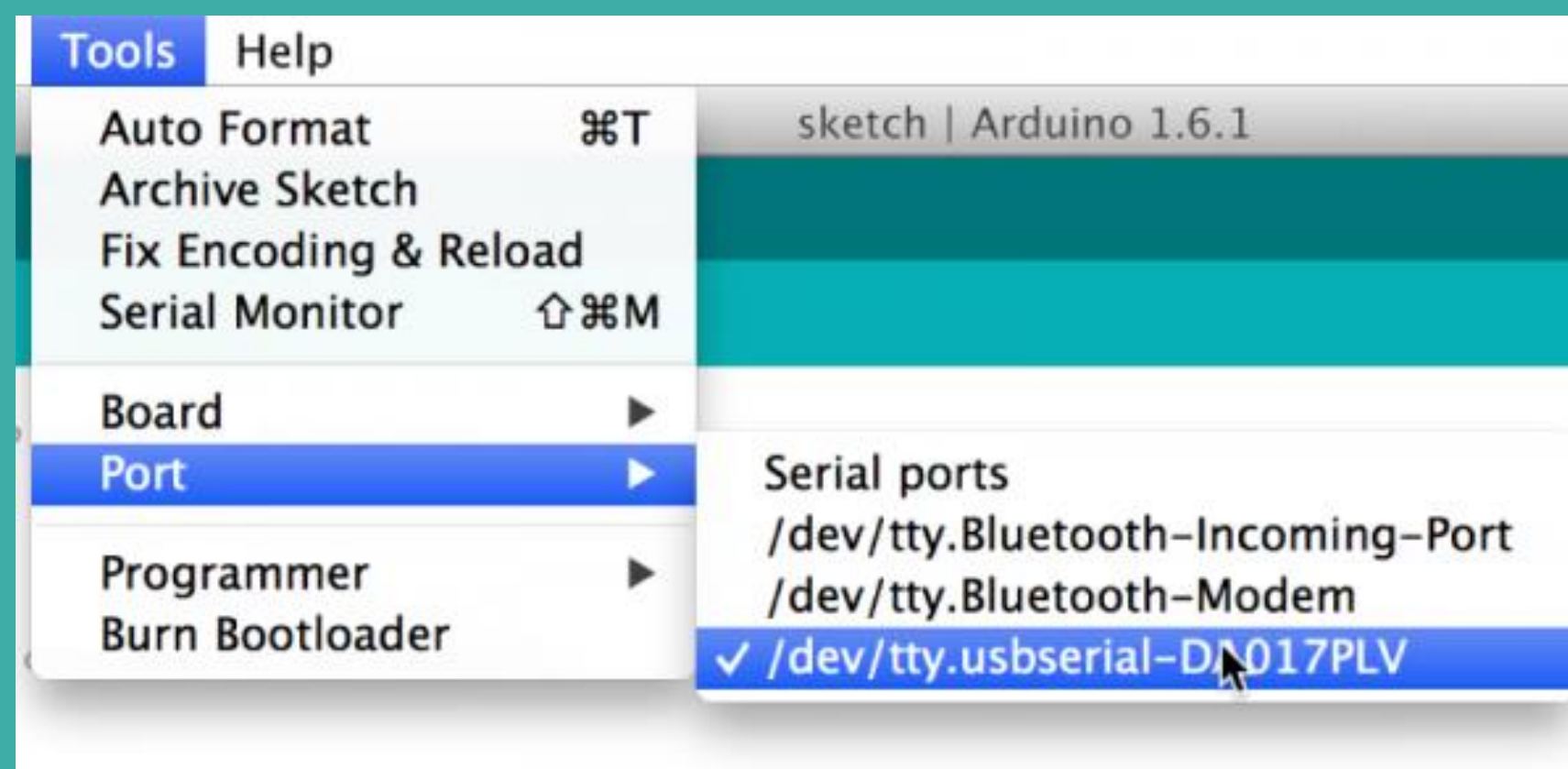
1. Connect your board to PC using USB



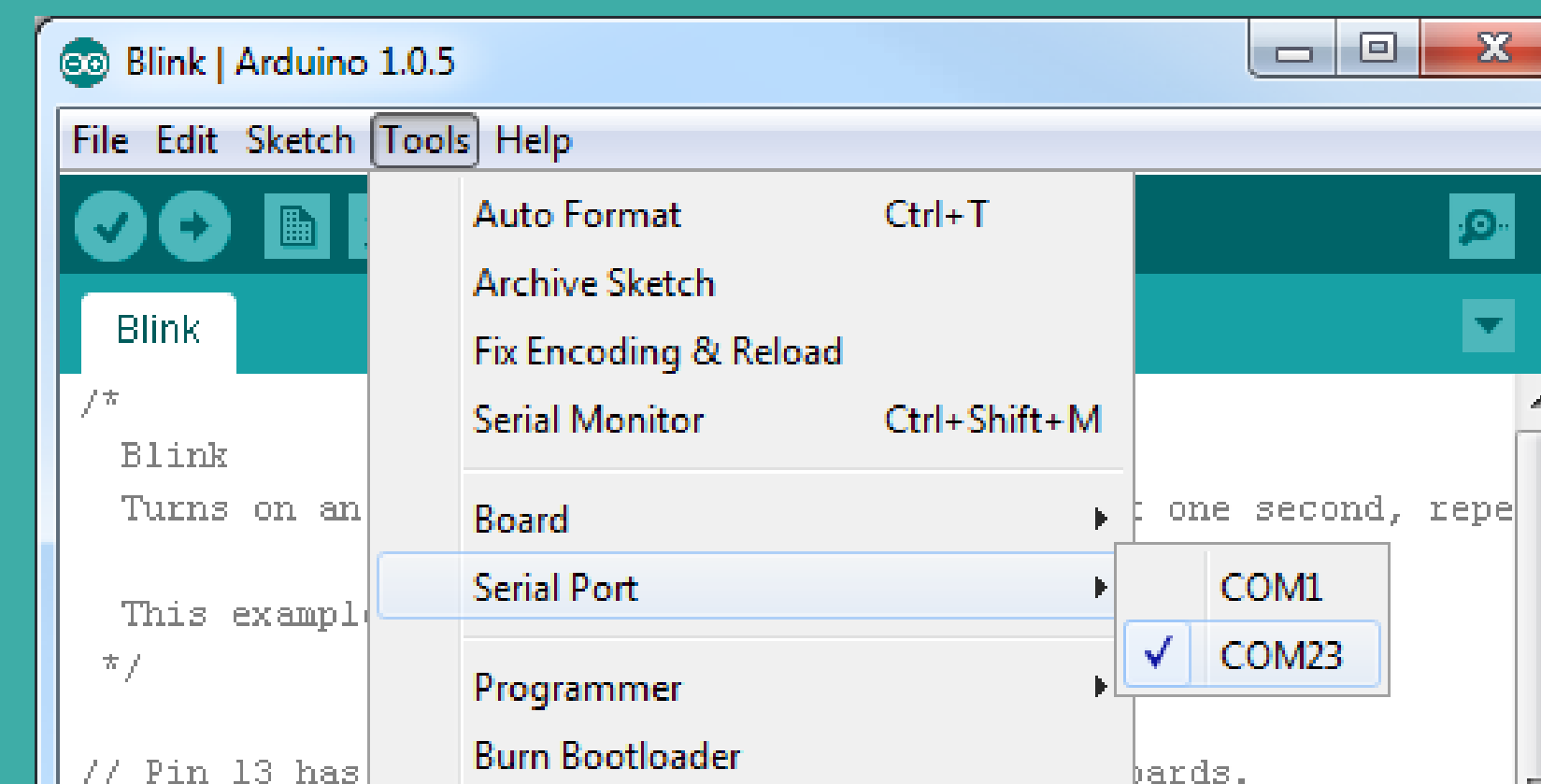
2. Select the board



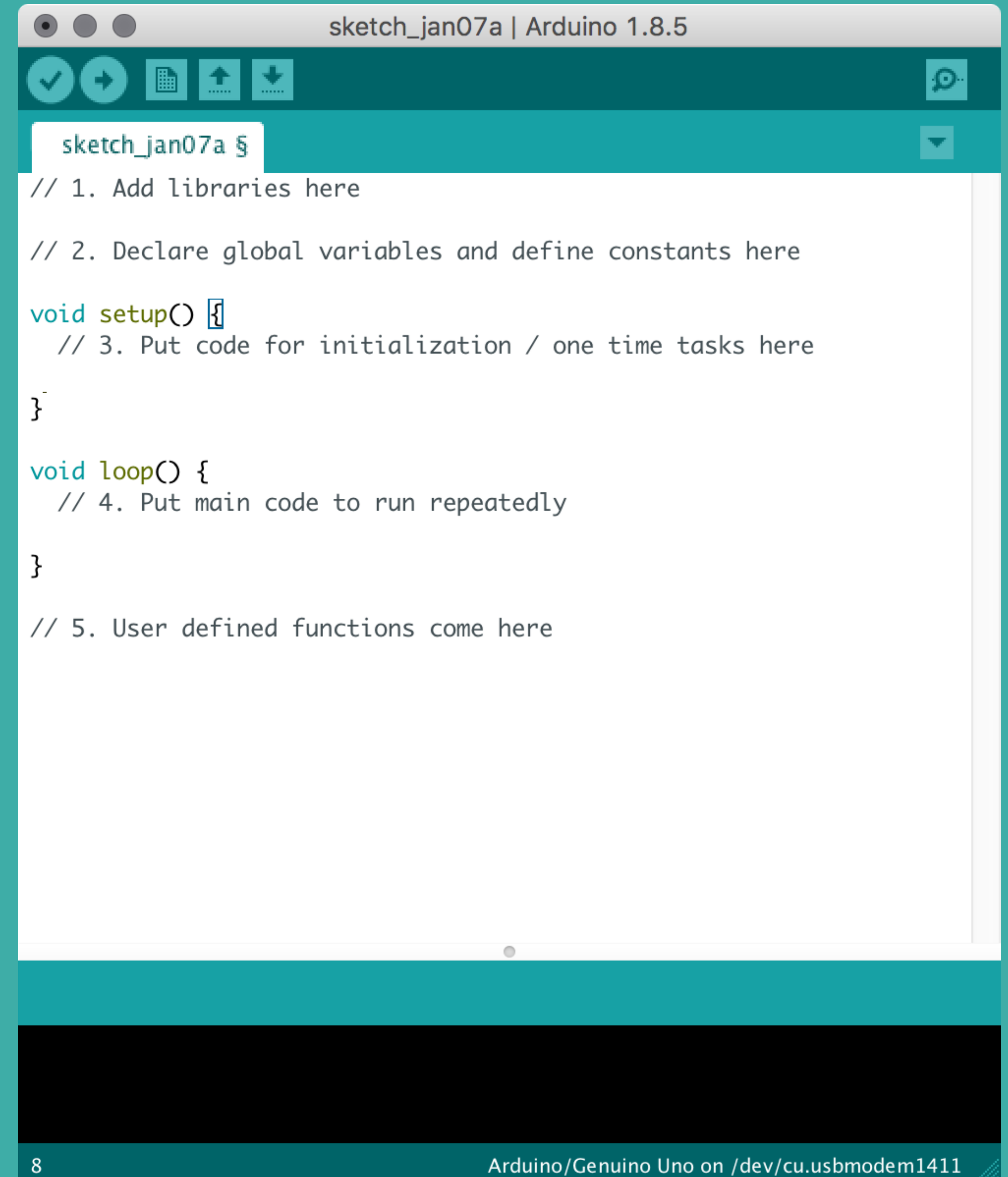
3. (MAC) Select the port



3. (Windows) Select the port



Code Basics



The screenshot shows the Arduino IDE interface. The title bar reads "sketch_jan07a | Arduino 1.8.5". The toolbar contains icons for checking, running, saving, uploading, and downloading. The code editor shows the following code:

```
sketch_jan07a $
// 1. Add libraries here

// 2. Declare global variables and define constants here

void setup() {
  // 3. Put code for initialization / one time tasks here
}

void loop() {
  // 4. Put main code to run repeatedly
}

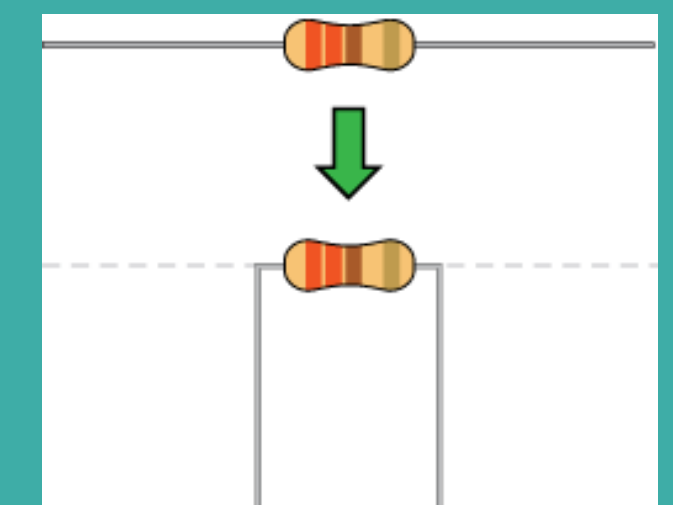
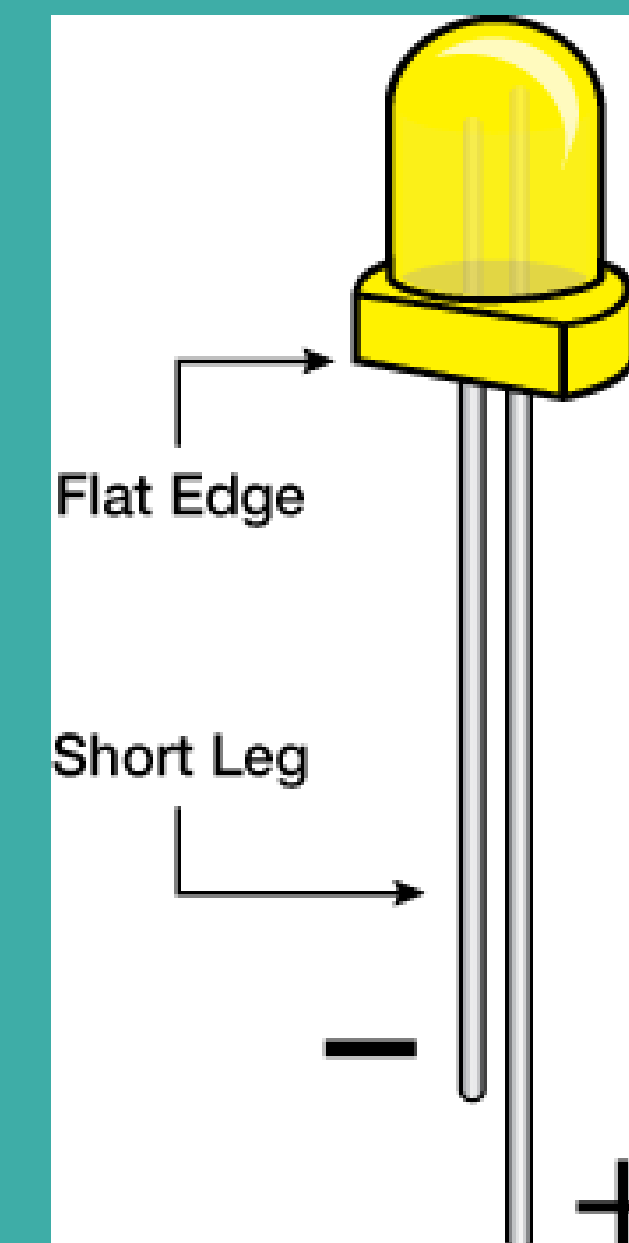
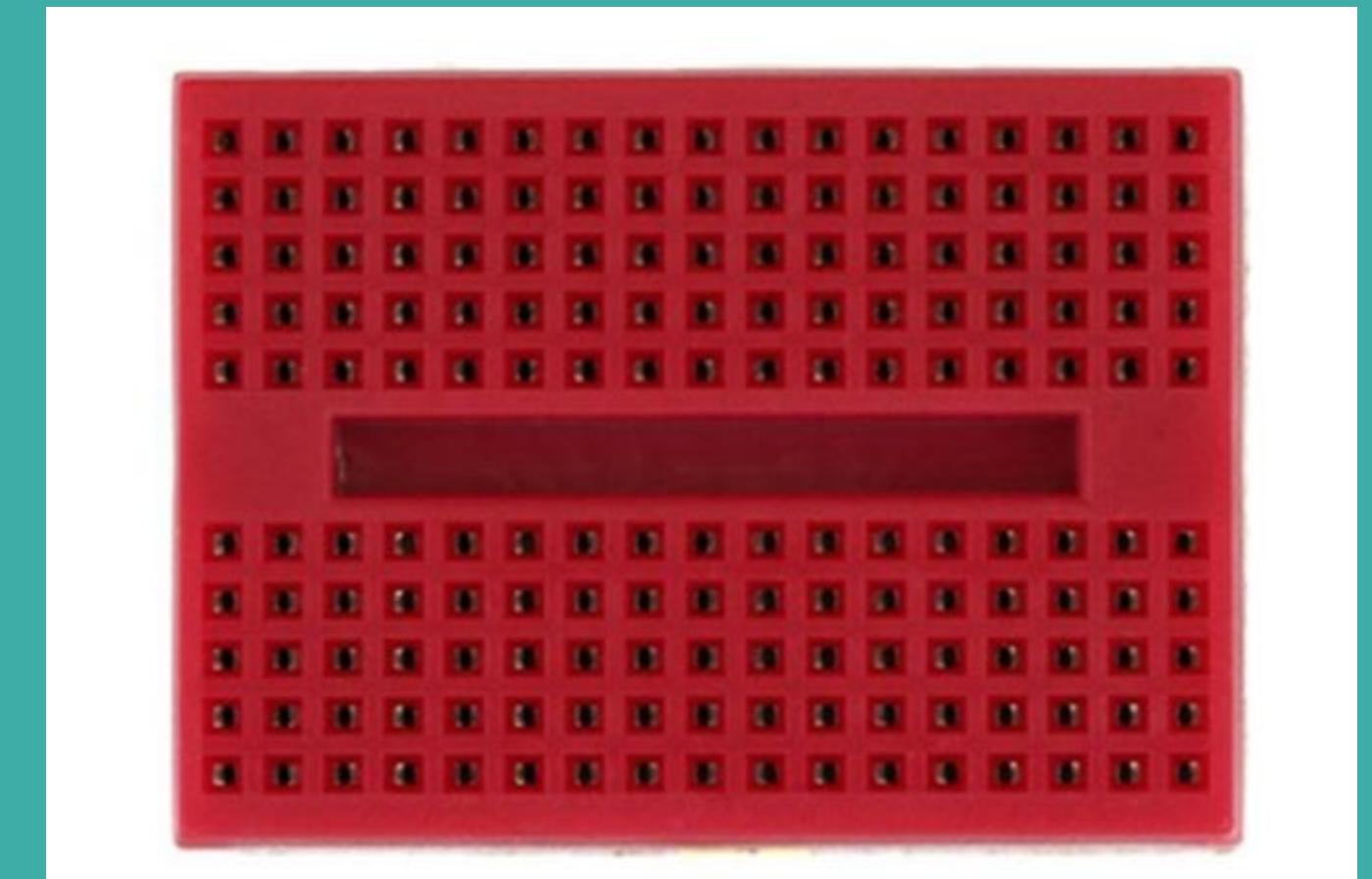
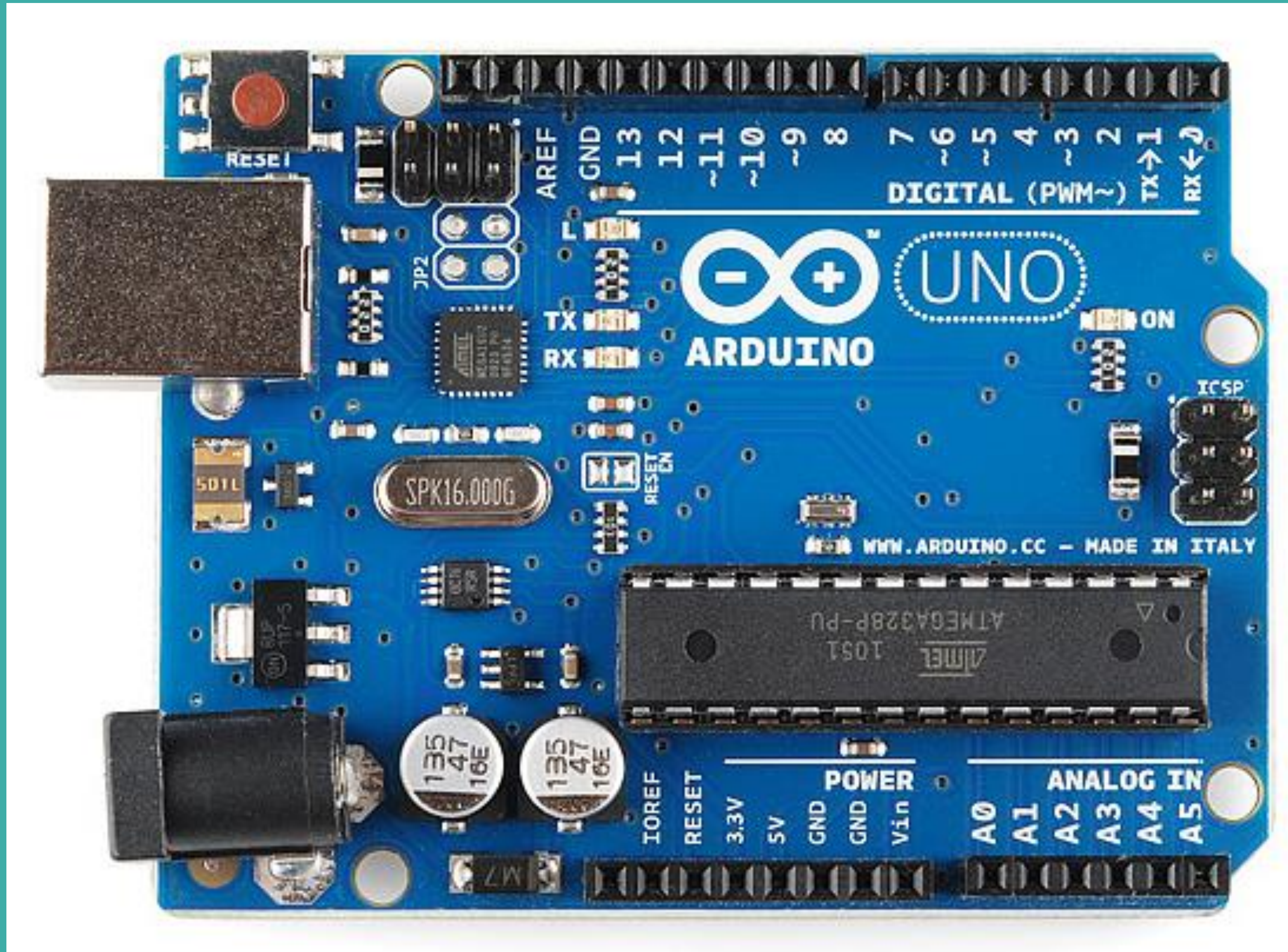
// 5. User defined functions come here
```

The bottom status bar shows "8" on the left and "Arduino/Genuino Uno on /dev/cu.usbmodem1411" on the right.

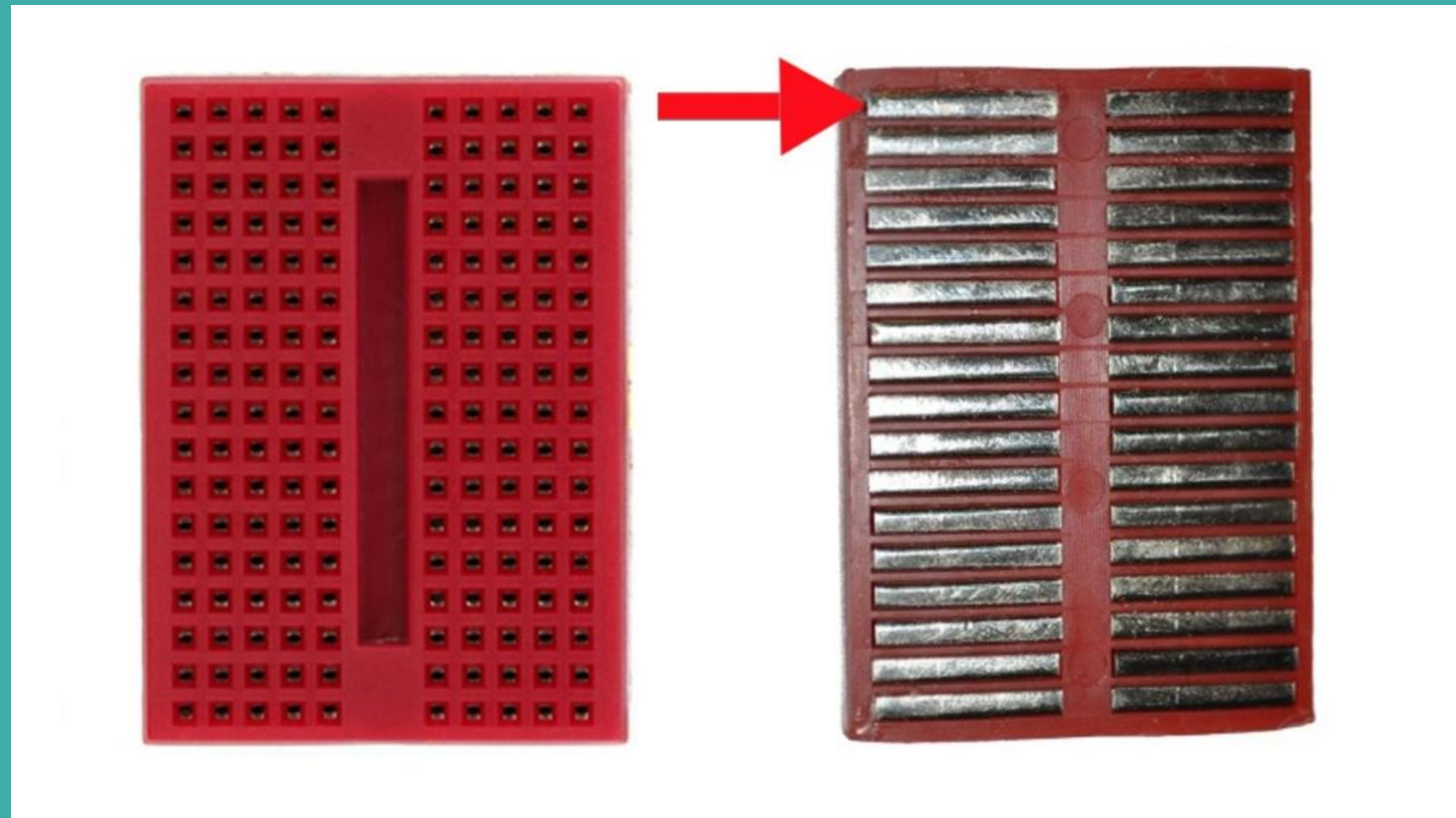
ACTIVITY 1

Blinking LED

Step 1: What You Will Need



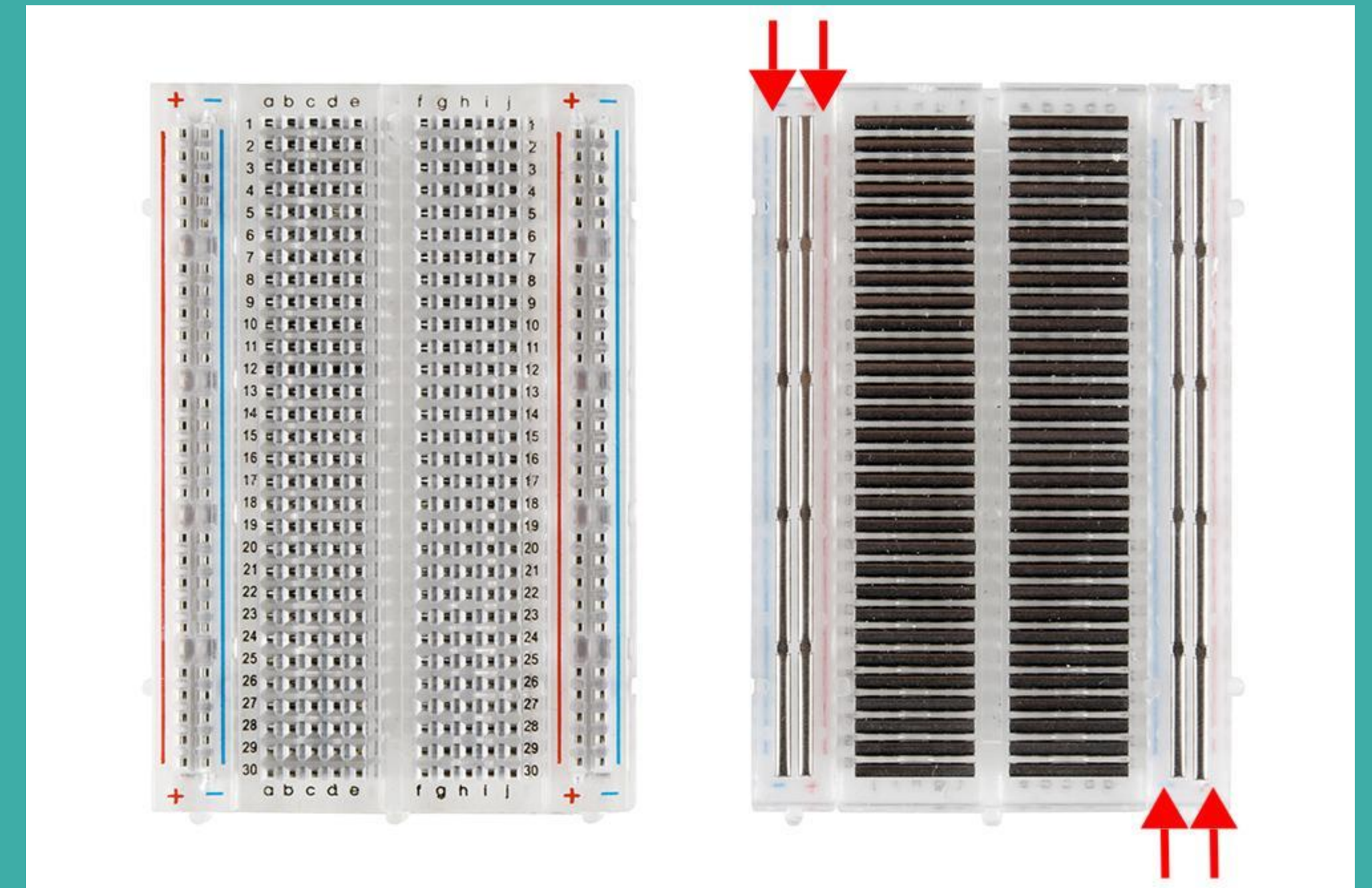
Breadboard Connections



Front view

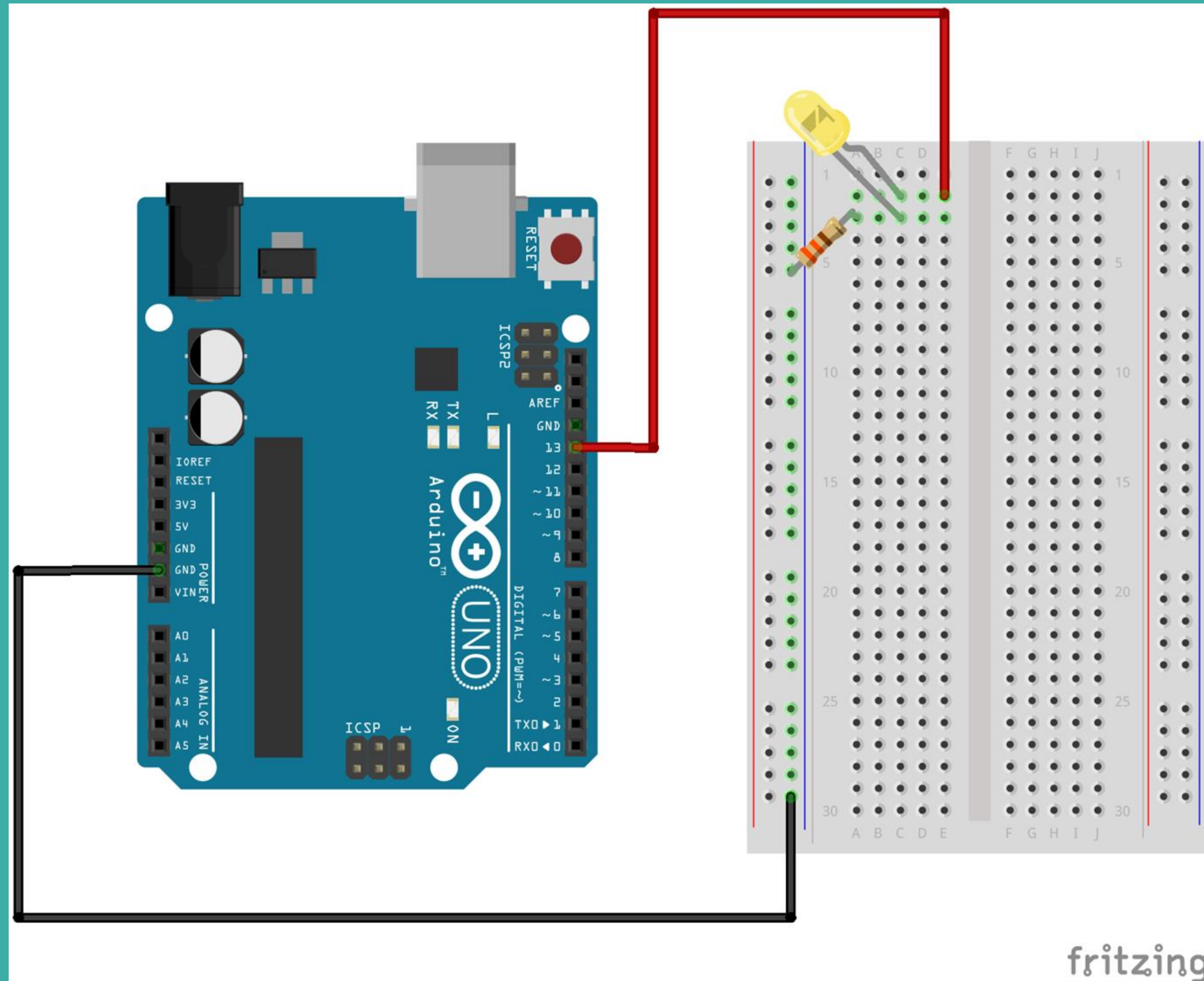
Rear view

Source: <https://cdn.sparkfun.com/assets/e/7/7/e/c/5175c500ce395f5a49000004.jpg>



Source: <https://cdn.sparkfun.com/assets/3/d/f/a/9/518c0b34ce395fea62000002.jpg>

Step 2: The Circuit



Step 3: The Code

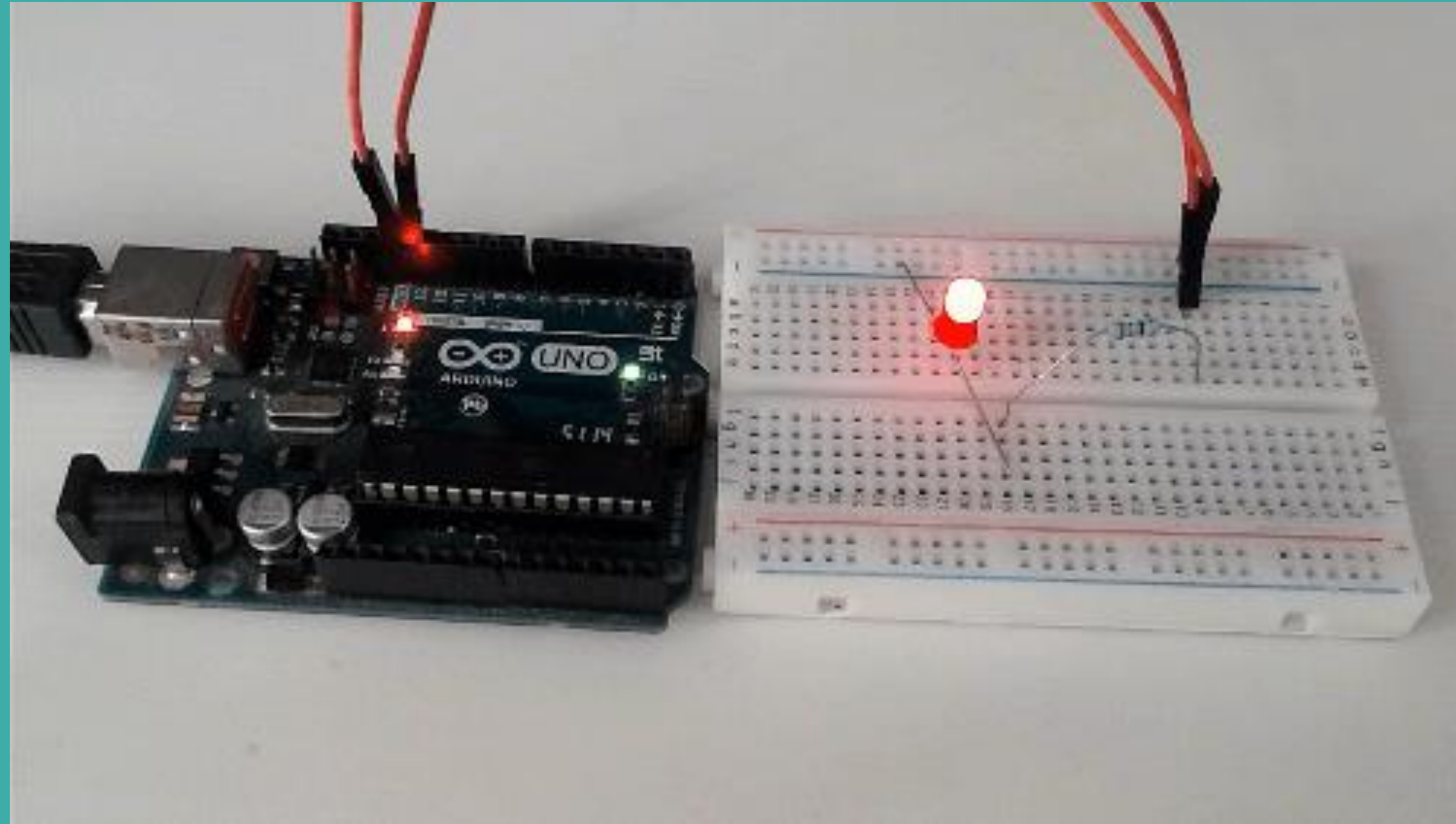


The screenshot shows the Arduino IDE interface with the 'Blink' example code loaded. The title bar reads 'Blink | Arduino 1.8.5'. The code editor contains the following text:

```
/*  
  Blink  
  
  Turns an LED on for one second, then off for one second, repeatedly.  
  
  Most Arduinos have an on-board LED you can control. On the UNO, MEGA and ZERO  
  it is attached to digital pin 13, on MKR1000 on pin 6. LED_BUILTIN is set to  
  the correct LED pin independent of which board is used.  
  If you want to know what pin the on-board LED is connected to on your Arduino  
  model, check the Technical Specs of your board at:  
  https://www.arduino.cc/en/Main/Products  
  
  modified 8 May 2014  
  by Scott Fitzgerald  
  modified 2 Sep 2016  
  by Arturo Guadalupi  
  modified 8 Sep 2016  
  by Colby Newman  
  
  This example code is in the public domain.  
  
  http://www.arduino.cc/en/Tutorial/Blink  
*/  
  
// the setup function runs once when you press reset or power the board  
void setup() {  
  // initialize digital pin LED_BUILTIN as an output.  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
// the loop function runs over and over again forever  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)  
  delay(1000); // wait for a second  
  digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW  
  delay(1000); // wait for a second  
}
```

The IDE interface includes a toolbar at the top with icons for checking, running, serial monitor, and file operations. The bottom status bar shows '20' on the left and 'Arduino/Genuino Uno on /dev/cu.usbmodem1411' on the right.

Expected Output

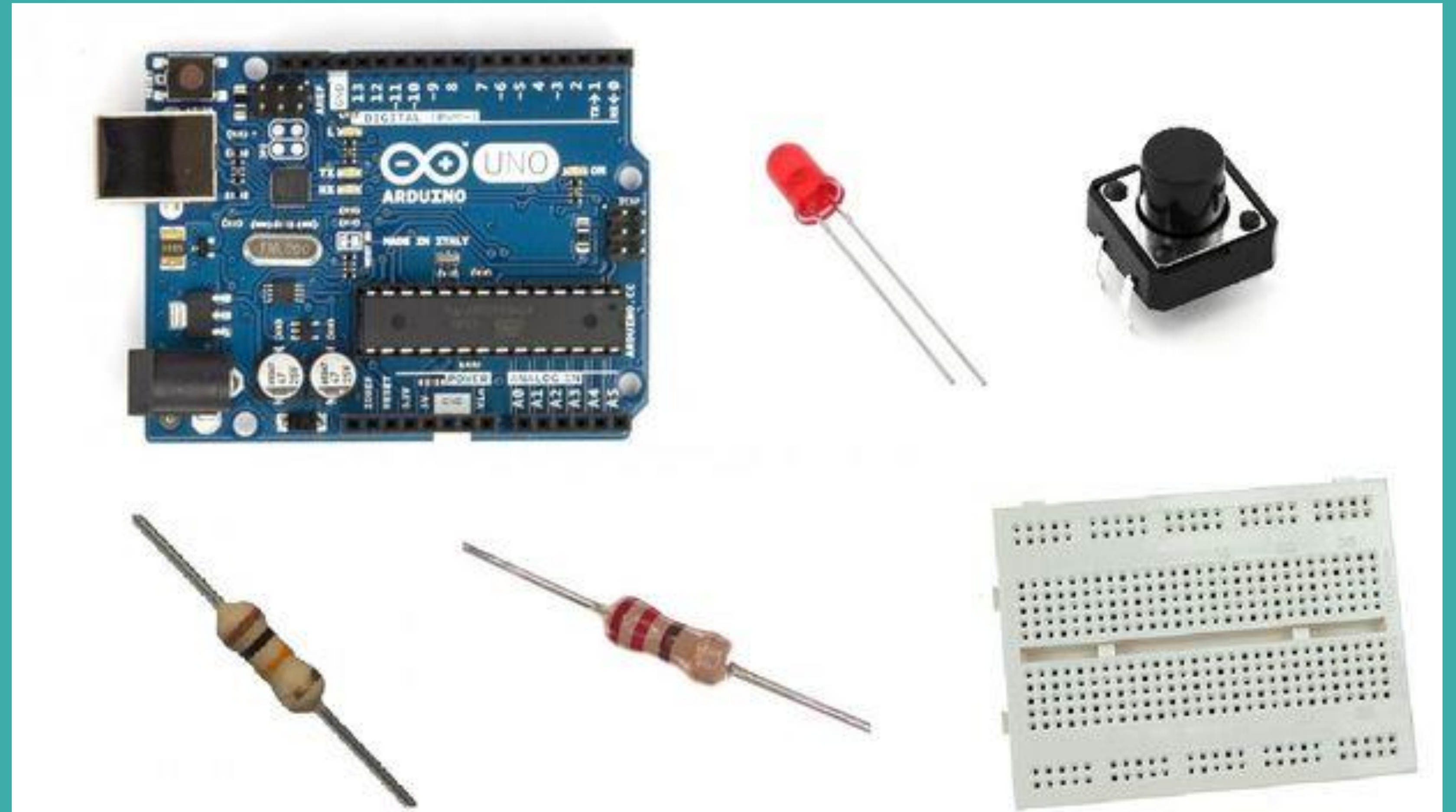


ACTIVITY 2

ON/OFF LED Using Pushbutton

Step 1: What You Will Need

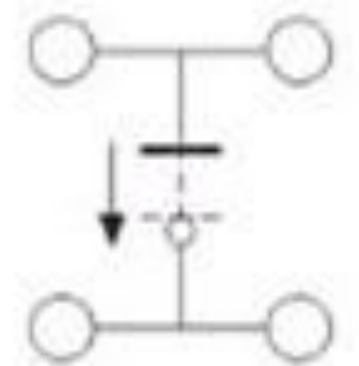
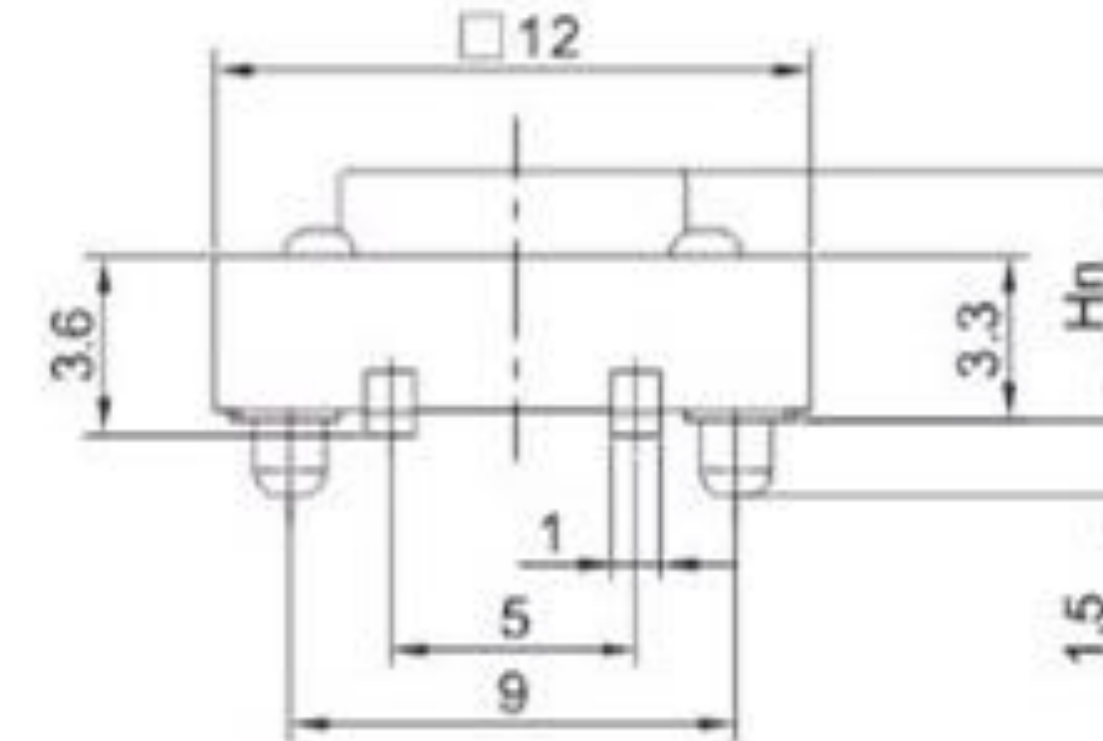
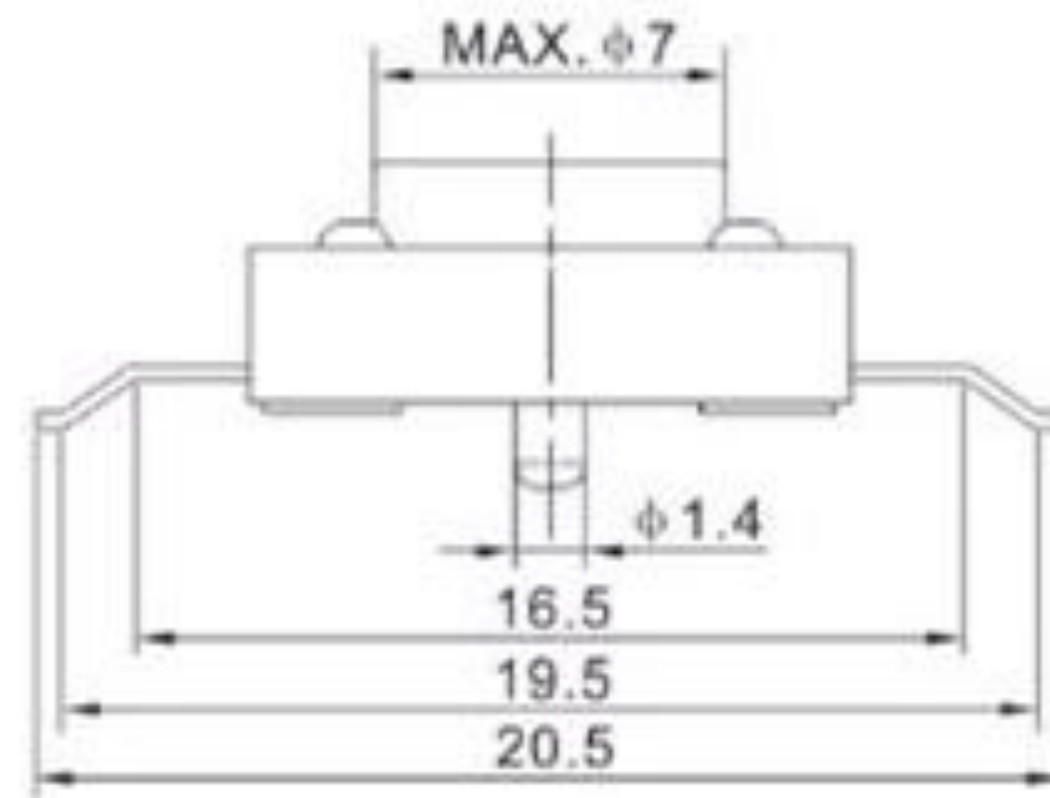
- Arduino uno
- Breadboard
- LED
- 220 Ohm & 10K Ohm resistor
- Push button



Push button Working



KFC-A07-PH5/D

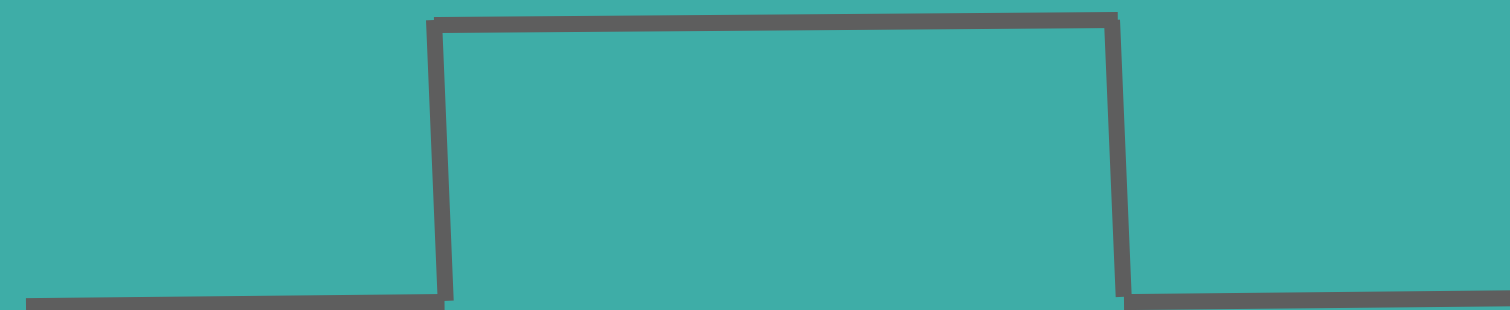


Be Aware

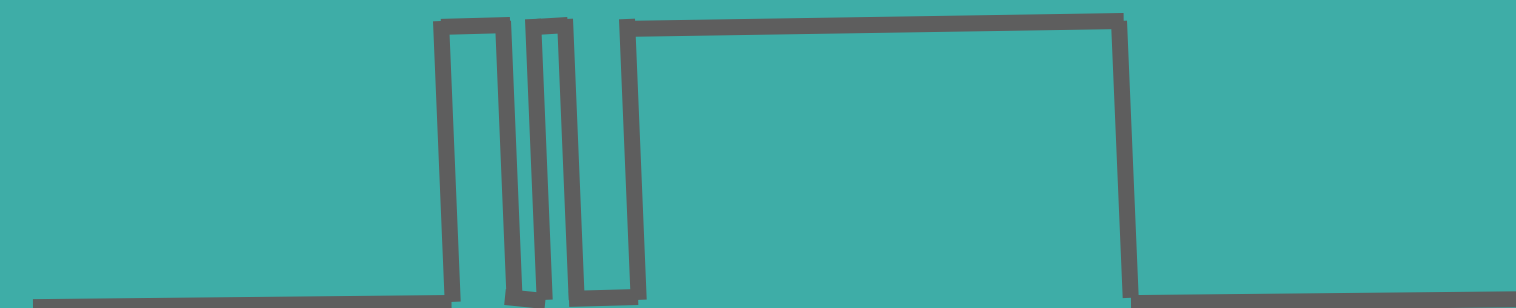


Bouncing/Debouncing

Expected
Behaviour

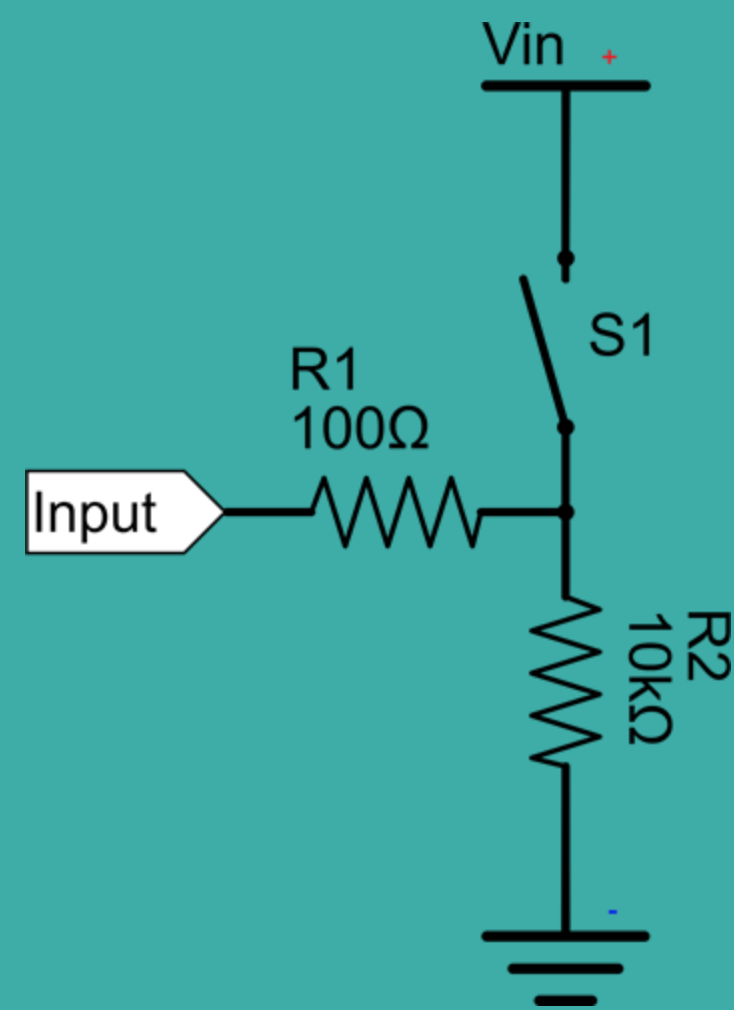


Actual
Switch



How can
you see
it?

Step 2: The Circuit



Pull down resistor

